CycleTwin

Synthetic cycle forecasting for clinician decision support

Digital Twin Challenge 2026 · Kushagra Gupta · VIT

Synthetic engineering prototype · no clinical validation

Not for diagnosis, contraception or conception planning.

The forecasting question

Can a system flag a likely delayed or absent ovulation early in a cycle, while showing uncertainty and limits?

- Target: synthetic anovulation or ovulation after cycle day 21
- Forecast points: cycle days 10, 14 and 18
- Decision-support framing; no diagnostic or reproductive-planning use

PMOS is a multi-system condition

Polyendocrine metabolic ovarian syndrome (PMOS; PCOS/PCOD)

The project frames the problem across reproductive and metabolic signals. The synthetic build currently evaluates only a reproductive-axis proxy.

Terminology follows Teede et al., The Lancet, 2026. The simulated phenotype labels are not diagnostic criteria.

Why a digital-twin framing

Planned design

- Reproductive follicle and ovulation dynamics
- Metabolic glucose dynamics
- Shared insulin-resistance latent with reverse cycle coupling

Current build

- Stochastic synthetic cycle generator
- Fixed risk score and temperature coverline heuristic
- No coupled physiological twin or glucose simulator

Data fusion and sensor tiers

Tier 0

Period logs

›

Tier 1

+ resting HR*

›

Tier 2

+ nightly temp

›

Tier 3

+ CGM windows

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Tier 4

+ EHR labs

Implemented evaluator: neutral period-only score; tier 1 uses resting-heart-rate offsets; tier 2 adds nightly temperature.

* Other wearable streams are generated but not fused. CGM and EHR tiers are not implemented.

Synthetic data and leakage control

- 1,200 synthetic patients × 6 cycles per generator seed
- Three fixed seeds: 17, 29, 43
- Patient-level split: 70% train, 15% validation, 15% held-out test
- No personal exports, credentialed data or external health records

The generator carries latent traits and noisy observations; the evaluator separates patients across partitions.

Forecast model

Risk score

- Tier 0: neutral baseline
- Tier 1: resting-heart-rate offset
- Tier 2: temperature rise relative to an early-cycle baseline

Ovulation-day estimate

- Temperature coverline heuristic
- Fallback: forecast day + 10
- No EKF, conformal interval or hormone model

All settings are simulator assumptions. No clinical interpretation is supported.

Safety and abstention

Planned safety layer

- Review for thyroid/prolactin flags
- Abstain for hormonal contraception and pregnancy-related states
- Abstain for age and sensor completeness conditions
- Show risk-coverage and subgroup coverage

Current status

These rules, coverage intervals and abstention logs are not implemented in G1. The dashboard is an XML parsing demo.

Synthetic held-out results

Tier 2 · forecast day 14 · held-out patients

Seed 97: AUROC 0.637 · AUPRC 0.598 · Brier 0.306 · event rate 0.506

Seed 101: AUROC 0.638 · AUPRC 0.604 · Brier 0.316 · event rate 0.523

Seed 103: AUROC 0.663 · AUPRC 0.615 · Brier 0.309 · event rate 0.511

Patient-cluster bootstrap 95% intervals for AUROC and AUPRC are in results.json. Synthetic generator performance only.

No comparison with real-world published accuracy is made.

Illustrative counterfactuals

Planned scenario inputs

- Regular 7–8 hour sleep
- Daily 30-minute walk
- Lower-glycaemic-index Indian food swap
- Lower evening stress
- Illustrative 5% weight change

Current status

No counterfactual simulator or intervention effects are implemented. Future effect ranges would be assumptions, not treatment advice.

Planned label: illustrative assumption, not a treatment recommendation.

Robustness and failure modes

- Current robustness sweeps are not implemented
- No wearable dropout, fever, travel or shift-work stress tests
- No simulator-shift experiment
- Known limitation: generator labels and signals share the same simplified assumptions

The synthetic benchmark cannot establish performance under real sensor noise or clinical variation.

Dashboard and Apple Health input

- Streaming XML parser for a limited HealthKit export subset
- Synthetic export writer; upload the file in the Streamlit demo
- Canonical daily records include wrist temperature, resting HR, HRV, steps and period starts
- CSV adapter supports date, temp, rhr, hrv, steps, period_start

The synthetic export is a schema-compatibility smoke test, n=1; it clinically validates nothing.

Related work

- Teede et al. · PMOS naming consensus · The Lancet, 2026
- Teede et al. · 2023 international guideline · JCEM
- Lin et al. · cycle glucose variation using CGM · npj Digital Medicine, 2023
- Collins et al. · TRIPOD+AI reporting guidance · BMJ, 2024

These works inform terminology, future measurement choices and reporting. They do not validate this simulator.

Limits and next steps

- Add coupled reproductive and metabolic dynamics with a shared latent
- Implement abstention, split conformal intervals and the alert-history budget
- Add CGM/EHR tiers, counterfactuals and robustness tests
- Seek clinician co-design and real-data validation before any clinical use

Synthetic engineering prototype · no diagnosis · no treatment or reproductive planning advice

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